

Practice exam 1

Units 3 & 4 Paper 2

Perusal time: 5 minutes

Working time: 90 minutes

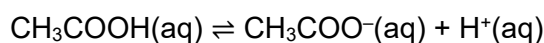
Section 1: Short response

Instructions

- If you need more space for a response, use the additional pages at the back of the book.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. “See page ...”
 - If you do not do this, your original response will be marked.
- This paper has **10 questions** and is worth **68 marks**.

QUESTION 1 (5 marks)

When 0.3 mol of the weak acid acetic acid is added to 1.0 L of water, the resulting solution has a pH of 3.
The equation for its ionisation is:



- a** Determine the expression for the acid dissociation constant, K_a , for this reaction. [1 mark]

- b** Calculate the concentration of H^+ in the solution. [1 mark]

- c Calculate the concentration of the undissociated acetic acid remaining in the solution. [1 mark]

- d Calculate the value of the acid dissociation constant, K_a . [1 mark]

- e Explain the meaning of the term “weak acid”. [1 mark]

QUESTION 2 (9 marks)

A student sought to determine the concentration of ethanoic acid in vinegar by performing a titration with sodium hydroxide. They recorded the following data in their logbook.

Concentration of NaOH(aq)	0.14 M
Volume of undiluted vinegar	5.00 mL
Total volume of diluted vinegar	50.00 mL
Volume of diluted vinegar used in each titration	10.00 mL
Titres of NaOH	7.20 mL, 7.60 mL, 7.25 mL, 7.30 mL

- a Determine an equation for the reaction taking place in this titration. [1 mark]

- b** Calculate the number of moles of base (sodium hydroxide) used in the average titre. *[2 marks]*

- c** Calculate the concentration of ethanoic acid in the original vinegar solution. *[4 marks]*

- d** Sketch the expected titration curve for this titration and suggest an indicator that can be used if the equivalence point occurs at pH 8.0. *[2 marks]*

QUESTION 3 (6 marks)

A chemist wishes to make ethyl propanoate, a major component in the odour of pineapple.

- a** Draw the structure of the ester ethyl propanoate. [1 mark]

- b** Using ethanol and propan-1-ol as starting materials, draw a scheme to show how ethyl propanoate would be prepared. Include the names of all reagents, catalysts and/or conditions required. [3 marks]

- c** Identify the names for the reactions in part b. [2 marks]

QUESTION 4 (8 marks)

A student wishes to construct a galvanic cell using 1.0 M solutions of AgNO_3 and PbCl_2 , a salt bridge and pieces of silver and lead metal.

- a** Draw an arrangement of the galvanic cell using these materials. Your answer should indicate the polarity of the electrodes and the metals that correspond to the anode and cathode. *[4 marks]*

- b** Identify the reactions occurring at each electrode *[2 marks]*

- c** Determine the overall cell equation. *[1 mark]*

- d** Explain the purpose of a salt bridge in a galvanic cell. *[1 mark]*

QUESTION 5 (4 marks)

Hydrogen sulfate ion (HSO_4^-) is amphoteric.

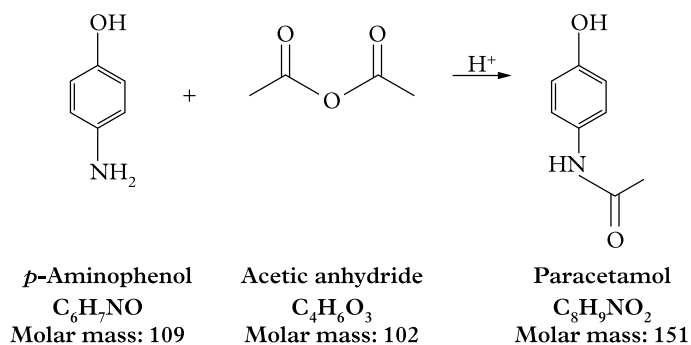
- a** Explain the meaning of the term “amphoteric”. [1 mark]

- b** Identify two chemical equations to show HSO_4^- acting as an amphoteric substance in aqueous solution. [2 marks]

- c** Hypobromous acid (HBrO) is a weak acid with an acidity constant, $K_a = 2.0 \times 10^{-9}$ at 25°C . Calculate the pH of a 1.0 M solution of hypobromous acid at this temperature. [1 mark]

QUESTION 6 (4 marks)

Paracetamol is a widely used painkiller. It is produced by the reaction of *p*-aminophenol and acetic anhydride, as shown below.



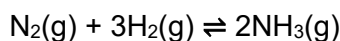
- a** If 5.0 g of *p*-aminophenol and 6.5 g of acetic anhydride are used in this reaction, determine which is the limiting reagent. [2 marks]

- b** Determine the amount of paracetamol that could theoretically be formed in this reaction. [1 mark]

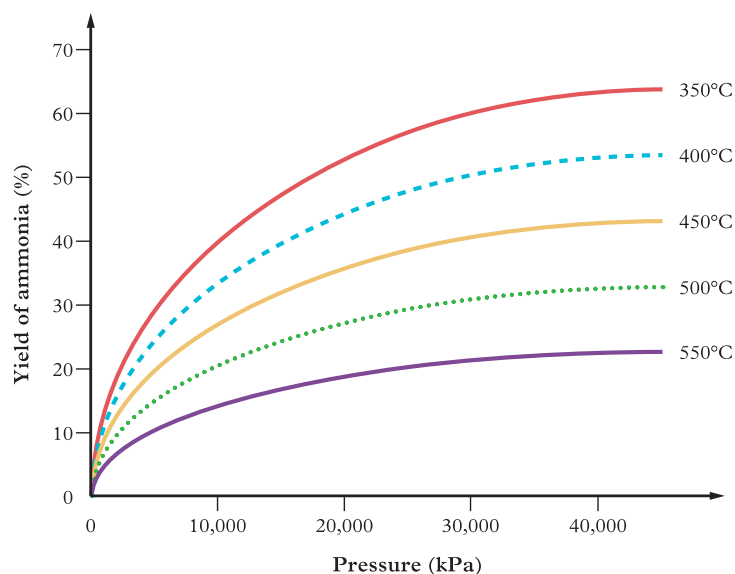
- c Calculate the percentage yield of paracetamol for this reaction if only 3.2 g of paracetamol was produced. [1 mark]

QUESTION 7 (5 marks)

Ammonia (NH₃) exists naturally in the environment and the human body, and is manufactured on an industrial scale in the Haber process.



The graph below shows the yield of ammonia formed at different pressures and temperatures.



- a Using the graph, determine the approximate yield of ammonia at a pressure of 30,000 kPa and temperature of 400°C. [1 mark]

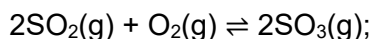
- b Based on the graph, explain whether the production of ammonia is exothermic or endothermic. [2 marks]

- c The reaction conditions used in the production of ammonia in the Haber process are 20 MPa and 400–500°C. Explain why lower temperatures are not used in this reaction. [1 mark]

- d List two uses of ammonia. [1 mark]

QUESTION 8 (4 marks)

One of the key reactions in the contact process is the conversion of sulfur dioxide (SO₂) to sulfur trioxide (SO₃) on reaction with oxygen (O₂), according to the equation shown below:



$$\Delta H = -197 \text{ kJ mol}^{-1}$$

Explain the effect on the position of equilibrium and the value of the equilibrium constant if:

- a the temperature of the reaction is increased [2 marks]

- b oxygen is added [2 marks]

QUESTION 9 (17 marks)

Compounds that belong to a homologous series have the same functional group and similar chemical properties.

a Identify the homologous series that the compounds in the following table belong to. [3 marks]

	Name of homologous series
$ \begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	
$ \begin{array}{c} \text{H}_3\text{C} \quad \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \quad \text{H} \end{array} $	
$ \begin{array}{cccc} & \text{H} & \text{H} & \\ & & & \\ \text{H}-\text{C}\equiv\text{C}-\text{C}-\text{C}-\text{H} \\ & & & \\ & \text{H} & \text{H} & \end{array} $	

b Explain why hydrocarbons have lower melting and boiling points than other organic compounds of similar molar mass. [2 marks]

c Determine the chemical equation for the reaction that occurs between butane and chlorine in the presence of ultraviolet light to produce a monohaloalkane and identify the name of this reaction. [2 marks]

- d Determine the structures and names of all the possible structural isomers of the organic product in the reaction in part c. [8 marks]

- e Explain why mass spectrometry would not be a suitable technique to identify the isomers formed in this reaction. [2 marks]

QUESTION 10 (6 marks)

A peptide chemist synthesised the tripeptide Gly-Phe-Val.

- a Draw the structure of this tripeptide and identify all of the peptide bonds. [2 marks]

- b** Draw the structure of the tripeptide as a zwitterion and at pH 2. [2 marks]

- c** If the tripeptide were a part of a protein, describe the type of interactions you would expect in this region of the protein. [2 marks]

END OF PAPER

